DDA2020 final exam

Fundamentals of Machine Learning (香港中文大學)

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DDA2020 Machine Learning
Final Exam

1 Multiple-choice questions (2 points per question) (30 points)

Note that there might be one or more correct option(s). If there are more than one correct answer, you should select all the correct options in order to get full marks. If your answer is correct but incomplete, you will only get partial marks. If any incorrect option is chosen, you will get zero mark.

1.1 Which one(s) are the sub-area of artificial intelligence?

A. Computer Vision
B. Robotics
C. Natural Language Processing
D. Machine Learning
E. Optimization

1.2 Suppose in a 2-class classification problem, our standard logistic regression model achieves 99% accuracy on the training set, but only 51% accuracy on the test set. Which of the following modifications might potentially improve our algorithm's test accuracy?

A. Use regularized logistic regression.
B. Use polynomial hypothesis function to replace the original hypothesis function $\mathbf{w}^\top \mathbf{x} + b$.
C. Add more training data.
D. Add more testing data.

1.3 Which of the following statement(s) is/are correct?

A. Backpropagation consists of a forward pass that computes the error, and a backward pass that adjusts the weights.
B. In a feedforward neural network, the information always moves in one direction.
C. A convolutional neural network is a fully connected network.
D. When training a multi-layer perception, we compute the error using a loss function at the output layer and pass the gradients from the output layer backwards to the input layer to update the weights.

1.4 Which of the following statement(s) is/are correct?

A. Toss a coin 4 times, then there will be 4 possible events.
B. If the probability of Event A is 0.1 and the probability of Event B is 0.01, then information of A is smaller then B.
C. KL divergence is positive and non-symmetric.
D. Both Binomial and Gaussian distributions are continuous distributions.

1.5 Which of the following statement(s) is/are correct?

A. When the number of training set n is very large and the feature dimension d is very small, we prefer to use the closed form solution rather than gradient descent method.
B. Linear regression can only be used for regression tasks.
C. The decision boundary of Ridge regression is non-linear in the original feature space.
D. The decision boundary of polynomial linear regression is non-linear in the original feature space.
E. If standard linear regression is overfitting, we can try Ridge regression.
F. If standard linear regression is overfitting, we can try polynomial linear regression.

1.6 Which of the following statement(s) is/are correct?

A. Standard SVM cannot handle a non-linearly separated data.
B. SVM with slack variables can handle a non-linearly separated data and its training error could be 0.
C. Kernel SVM can perfectly fit non-linear separated data.
D. Margin is the distance from the closest point of positive and negative classes to the decision boundary

1.7 Which of the following statement(s) is/are correct?

A. PCA can give non-linear projection of data.
B. Dimensionality reduction tasks can be solved by supervised or unsupervised learning methods.
C. We project N dimensional data points to a k dimensional space, the dimension of reconstruction points is lower than that of the original data.
D. In PCA, we choose the k eigenvectors with the top k largest eigenvalues to form a new matrix.

1.8 Which of the following statement(s) is/are correct?

A. Decision tree can only handle training data with numerical attributes
B. When we set the minimal size of leaf node t[illegible] large value, then we will grow a deep decision

C. When we set the maximal depth to be a small value, then we will grow a shallow decision tree.

D. When we increase the number of decision trees, the training performance of Bagging will often be better.

1.9 Which of the following would you apply unsupervised learning to?

A. Given the data of body dimensions collected from 1,000 consumers, determine the size of the clothes to be produced.

B. Develop a model to predict the stock market.

C. Given a large dataset of medical records of patient including the disease name, predict the disease of the new patient by the symptom.

D. Compress a high-resolution image to a low-resolution image.

1.10 A class has 10 students. They received marks for their mid-term quiz as follows.

Student ID	01	02	03	04	05
Marks	90	50	66	71	82

Student ID	06	07	08	09	10
Marks	72	75	68	99	60

To group the students into two tutorial groups according to their marks, we use k-means. We pick student 03 as the initial centroid for Group A, and 07 for Group B, and assign the students to the two groups using Euclidean distance.

A. We will have 4 students in Group A.

B. We will have 5 students in Group B.

C. If we change the initial centroid, the clustering result by kmeans will not be changed.

D. With the above group assignments, we re-estimate the new centroids for the two groups. The new centroid of Group A is 60.

1.11 Which of the following statement(s) is/are correct?

A. When the TPR-curve is plotted as y-axis and the FPR-curve is plotted as x-axis, the plot is called the ROC curve.

B. $TPR + FPR = 1$.

C. Accuracy $= \frac{TPR+TNR}{2}$.

D. As the classification threshold increases, the FNR increases.

1.12 Which of the following statement(s) is/are correct?

A. When we increase the model complexity, bias drops and variance grows in the training set.

B. When we increase the model complexity, bias drops and variance grows in the testing set.

C. When we increase the model complexity, both bias and variance drops in the training set.

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D. When we increase the model complexity, bias grows and variance drops in the testing set.

1.13 Which of the following statement(s) is/are correct?

A. If we fix the covariance matrix $\mathbf{\Sigma}$ as the identity matrix $\mathbf{I}$ when fitting GMM using EM, then it is equivalent to the standard K-means algorithm.

B. It is guaranteed that EM algorithm can improve the log likelihood function.

C. The latent variable in latent variable models must be discrete.

D. Suppose that f is a concave function and X is a random variable, then $f(\mathbb{E}(X)) \geq \mathbb{E}(f(X))$.

1.14 Given a linear system $\mathbf{X}\mathbf{w} = \mathbf{y}$, where $\mathbf{w} \in \mathbb{R}^d$ is the variable we want to solve, and $\mathbf{X} \in \mathbb{R}^{m \times d}$ and $\mathbf{y} \in \mathbb{R}^m$ are the data provided, which of the following statement(s) is/are correct?

A. When $m = d$, it is guaranteed to obtain a unique solution.

B. When $m > d$, this linear system is called under-determined system, and there is no solution.

C. When $m > d$, this linear system is called over-determined system, and there is no solution.

D. When $m < d$, this linear system is called under-determined system, and there are infinite solutions.

E. When $m < d$, this linear system is called over-determined system, and there are infinite solutions.

1.15 Suppose we want to train a classifier in a supervised learning manner, in order to automatically evaluate student assignments. In this setting, which of the following statement(s) is/are correct?

A. The collected students assignments which have not been graded by teachers can be used as the experience.

B. The task is to predict the grades of the student assignments.

C. The accuracy of the predicted grades can be used as the performance measure.

D. The accuracy of the students' answers to the assignments can be used as the performance measure.

2 Calculations and Derivations (70 points)

2.1 (5 points) Consider two discrete random variables X and Y, $X \in \{0,1,2\}$, $Y \in \{2,3\}$. $P(Y=2) = 0.4$, $P(Y=3) = 0.6$. Given Y, X follows binomial distribution, *i.e.*, $P(X=k|Y=y) = \frac{n!}{k!(n-k)!}(\frac{1}{y})^k(1-\frac{1}{y})^{n-k}$, where $n = 2$.

1. Calculate the probability distribution of X, *i.e.*, $P(X)$. (1 point)
2. Calculate the conditional distribution of $P(Y$ 1). (Hint: using Bayes rule) (2 points)

3. Calculate $E(X)$, and $E(Y|X=1)$. (2 points)

2.2 (15 points) We define a CNN model as

$$f_{CNN}(\boldsymbol{X}) = Softmax(FC_1(Conv_2(MP_1(Relu_1(Conv_1(\boldsymbol{X})))))). \tag{1}$$

The size of the input data $\boldsymbol{X}$ is $36 \times 36 \times 3$; the first convolutional layer $Conv_1$ includes 10 $8 \times 8 \times 3$ filters, stride=2, padding=1; $Relu_1$ indicates the first Relu layer; MP_1 is a 2×2 max pooling layer, stride=2; the second convolutional layer $Conv_2$ includes 100 $5 \times 5 \times 10$ filters, stride=1, padding=0; FC_1 indicates the fully connected layer, where there are 10 output neurons; $Softmax$ denotes the Softmax activation function. The ground-truth label of $\boldsymbol{X}$ is denoted as t, and the loss function used for training this CNN model is denoted as $\mathcal{L}(y,t)$.

1. Compute the feature map sizes after $Relu_1$ and $Conv_2$ (2 points)
2. Calculate the number of parameters of this CNN model (hint: don't forget the bias parameter of in convolution and fully connection) (4 points)
3. Plot the computational graph (CG) of the forward pass of this CNN model (3 points) (hint: use $z_1, z_2, z_3, z_4, z_5, z_6$ denote the activated value after $Conv_1$, $Relu_1$, MP_1, $Conv_2$, FC_1, $Softmax$)
4. Based on the plotted CG, write down the formulations of back-propagation algorithm, including the forward and backward pass (6 points) (Hint: for the forward pass, write down the process of how to get the value of loss function $\mathcal{L}(y,t)$; for the backward pass, write down the process of computing the partial derivative of each parameter, like $\frac{\partial \mathcal{L}}{\partial w_1}, \frac{\partial \mathcal{L}}{\partial b_1}$)

2.3 (17 points) Derivations of SVM's objective function and solution according to dual problem. Given the training data $\{(\mathbf{x}_i, y_i)\}_i^n$, denoting the parameters as $\mathbf{w}, b$, the objective function of SVM is formulated as follows:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \tag{2}$$

$$\text{s.t. } y_i\left(\mathbf{w}^\top \mathbf{x}_i + b\right) \geq 1, \forall i$$

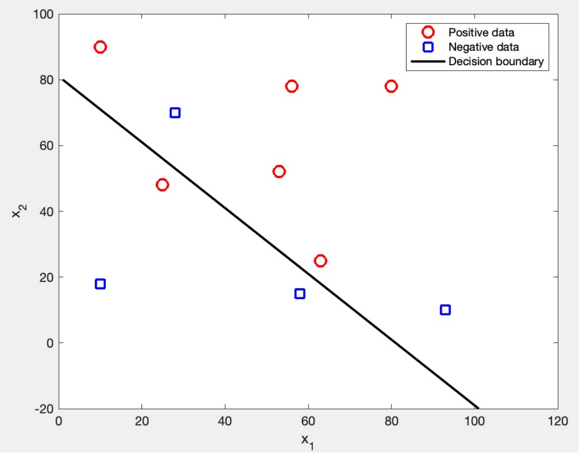
1. Derive the above objective from the perspective of large margin. (Hint: The margin is defined as the closest distance from the data point to the hyperplane. We wish to find a hyperplane that separate the data, meanwhile having the largest margin among all the hyperplanes. The distance of a point $\mathbf{y}$ to a hyperplane given by $\mathbf{f}_{\mathbf{w},b}(\mathbf{x}) := \mathbf{w}^T\mathbf{x} + b = 0$ is given by $\frac{|f_{\mathbf{w},b}(\mathbf{x})|}{\|\mathbf{w}\|}$) (6 points)
2. Derive the above objective from the perspective of hinge loss. (Hint: The objective function of SVM using hinge loss is given by

$$C\sum_i^m \mathbf{hinge\_loss}(\mathbf{x}, \mathbf{y}; \mathbf{w}, b) + \frac{1}{2}||\mathbf{w}||^2$$

You can firstly write down the explicit expression of hinge loss.) (5 points)

3. Derive the solution of $\mathbf{w}, b$ according to the Lagrangian function and KKT condition, and tell when a training data is called support vector. (Hint: firstly write down the Lagrangian function, then KKT conditions, then use the KKT conditions to derive the solution.) (6 points)

2.4 (10 points) Consider the figure below:



The red circles represent the data points in positive class and the blue squares represents negative. We construct a model to do the prediction and get a decision boundary as the black curve. The points on the top right of the decision boundary are predicted to be positive and other points at the bottom left of the boundary are predicted to be negative.

1. Write down the confusion matrix of this classification. (2 points)
2. Calculate the precision, recall and accuracy of this classification method. (3 points)
3. Calculate the FNR and FPR. (2 points)
4. Suppose that the posterior probabilities of the 6 positive data $p(y_i = +|\mathbf{x}_i)$ are 0.2, 0.5, 0.7, 0.7, 0.8 and 0.9, respectively. The posterior probabilities of the 4 negative data $p(y_i = +|\mathbf{x}_i)$ are 0.1, 0.3, 0.5 and 0.6, respectively. Calculate the AUC of this model. (3 points)

2.5 (15 points) Suppose that a random variable $\mathbf{x}$ follows the Gaussian mixture distribution with $p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. Where $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}, \boldsymbol{\pi} = \{\pi_1, \ldots, \pi_K\}, \boldsymbol{\mu} = \{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K\}, \boldsymbol{\Sigma} = \{\boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_K\}$ and latent variable $z \sim$ Categorical $(\boldsymbol{\pi})$, where $\boldsymbol{\pi} \geq 0, \quad \sum_k \pi_k = 1$.

1. **Likelihood Decomposition**. Show that

$$\ln p(\mathcal{D}; \boldsymbol{\Theta}) \geq \mathcal{L}(q; \boldsymbol{\Theta}), \forall \mathbf{q}, \boldsymbol{\Theta},$$

where

$$\mathcal{L}(q; \boldsymbol{\Theta}) = \sum_{n=1}^N \mathbb{E}_{q_n(z^{(n)})}\left[\ln\left(\frac{p(\mathbf{x}^{(n)}, z^{(n)};)}{q_n(z^{(n)})}\right)\right],$$

and $\mathbf{q}(\mathbf{z}) = \prod_{n=1}^N q_n\left(z^{(n)}\right)$. And, write down the gap between $\ln p(\mathcal{D}; \boldsymbol{\Theta})$ and $\mathcal{L}(\mathbf{q}; \boldsymbol{\Theta})$. (6 points)

2. **E-Step Derivation**. With given $\boldsymbol{\Theta} = \{\boldsymbol{\pi},$ [illegible] update $\mathbf{q}(\mathbf{z})$. (3 points)

3. **M-Step Derivation** Given $\mathbf{q}(\mathbf{z})$, update $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}$. (6 points)

2.6 (8 points) Consider a binary classification with the following training data. Each data is described by 3 attributes: Color, Size, Shape.

Example	Color	Shape	Size	Class
1	Red	Square	Big	+
2	Blue	Square	Big	+
3	Red	Square	Big	+
4	Red	Circle	Big	+
5	Green	Circle	Big	-
6	Blue	Circle	Small	-
7	Blue	Circle	Small	-
8	Green	Square	Big	-

1. Use the entropy method to find the best attribute for the root node. (Hint: Calculate the entropy and information gain) (4 points)

2. Build a complete tree according to the entropy method. The tree should include the class label and training data in each node. (2 points)

3. Compute the classification error of each node of the tree. (2 points)